Macao Tourism Recovery

Visitor composition, daily arrivals and non-gaming expenditure

CISC7204 Data Science and Data Visualization
Class C - Team 05
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Semester Project Proposal

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Project period: 5 November to 2 December 2026. All project activities are scheduled within four weeks of the proposal submission date.

AI acknowledgment: OpenAI Codex assisted with source checks, course mapping, code, charts and editing. The team will verify the calculations and interpretations and remains responsible for the submitted work. Values come from the cited official sources; generated text does not replace observations (OpenAI, 2026; University of Macau, 2026b).

Research overview

We propose to investigate whether Macao tourism recovered in the same way across total arrivals, daily distributions, visitor composition and non-gaming expenditure. Extending Assignment 2, the project combines 3,287 observed dates from 2017 to 2025, 55,879 date-by-source records, and 16 expenditure quarters covering 2019 and 2023-2025. Two notebooks already implement the proposed analyses. Their preliminary results establish feasibility; the four-week plan covers team verification, interpretation, reporting and final delivery.

Preliminary results show that 2025 total arrivals were 1.68% above 2019, while the daily median was slightly lower, P90 and IQR were higher, and the overnight share fell by 5.99 percentage points. Ridge was selected in 2024 validation, while OLS with a trend had lower 2025 test error. Official nominal spending per visit was about 23.00% above 2019 but 7.28% below 2024. These findings motivate a study of how recovery depends on the measure, distribution, composition and comparison period.

The project uses taught methods: data wrangling, descriptive statistics, empirical cumulative distributions, stratified comparisons, multiple linear regression, Ridge and model evaluation (University of Macau, 2026c, 2026g, 2026h, 2026k, 2026l, 2026m). Its depth comes from checking measurement definitions, joining data at compatible levels, comparing models honestly and explaining errors within the evidence.

Purpose and learning outcomes

By completion, the team should be able to check official data and observation units with Pandas; explain differences behind totals using distributions, proportions and stratified plots; compare OLS and Ridge with clear training, validation and test boundaries; and communicate reproducible findings and limitations. Tourism and economic measurement knowledge will guide the distinction between visits and people, overnight visitors and hotel guests, and nominal spending and real purchasing power. Each member will review evidence outside their main work package and explain one finding and its limitation during rehearsal. These outcomes develop the course skills of data analysis, visualization and critical interpretation (University of Macau, 2026a).

Research questions

ID	Question	Evidence
RQ1	How did total arrivals, overnight/same-day composition and source composition change between 2019 and 2025?	Annual arrivals, visitor-type shares, source shares and absolute changes
RQ2	How did the typical level, spread and upper quantiles of daily arrivals change, and did changes vary by month and weekday?	Mean, median, IQR, P90, ECDF and calendar strata
RQ3	How did calendar models selected using 2024 validation perform in 2025, and how did trend and Ridge affect errors?	Baseline, OLS and Ridge; chronological validation, test metrics and high-arrival errors
RQ4	Did arrivals, nominal non-gaming expenditure and official spending per visit recover together under same-quarter and annual comparisons?	Same-quarter 2019 indices; 2025 versus 2019 and 2024 annual comparisons; type shares and means

RQ1 and RQ2 describe observed historical changes. RQ3 evaluates estimation errors in a later period. RQ4 extends the original spending question and compares overall and visitor-type means. Associations and arithmetic contributions to change are not interpreted as causal effects of policies, visitor types or source markets.

Data sources and scale

Dataset	Coverage and size	Use
DSEC tourism statistics database	1 January 2017 to 31 December 2025; 3,287 dates	Daily totals and visitor types
Source breakdown from the same database	17 mutually exclusive document-issue groups; 55,879 records	Source composition and grouped comparisons
DSEC visitor expenditure survey	16 quarters in 2019 and 2023-2025	Total and type-specific spending, including spending per visit

Geographic region: Macao Special Administrative Region, China. Domain: tourism, specifically visitor arrivals, visitor composition and non-gaming visitor expenditure. The source breakdown is part of the arrivals database; the expenditure survey is the second dataset.

Two datasets are available through https://www.dsec.gov.mo/TourismDBWeb/ and https://www.dsec.gov.mo/en-US/Statistic?id=404. They provide administrative arrival counts and survey-based expenditure estimates, respectively (Statistics and Census Service, n.d.-a, n.d.-b). Daily breakdowns are available from 2017 (Statistics and Census Service, 2017). The full source archive retains official responses, request details, quarterly publications and extraction locations; the analysis package supplies the fixed input tables and source URLs.

The 55,879 source-group records share 3,287 dates and are not independent time observations. The periods 2017-2019 and 2023-2025 contain 2,191 dates and 37,247 source-group records. The 1,096 dates in 2020-2022 remain in the full history and are marked separately. Regression uses the 1,096 dates in 2023-2025.

Arrivals count visits, so one person may appear more than once. Source groups follow the place of travel-document issue, not nationality or residence. Overnight arrivals are not hotel guests or room nights. Expenditure is published from a quota survey and extrapolation; respondent records, survey weights and usable variance estimates are unavailable. All expenditure is in current-price Macao patacas (MOP) and excludes gaming.

Data preparation and quality checks

Source data remain unchanged. Daily total and visitor-type tables use date as the unique key; source records use date and source code. We check continuous dates, duplicate keys, non-negative counts, missing values and component sums, then use Pandas groupby, merge and pivot. Source codes are categories, not numerical distances. The wrangling workflow follows Module 02 (University of Macau, 2026j).

All 3,287 dates are present. Each date has 17 source groups whose sum equals total arrivals. The same-day count is missing on 16 September 2018; its raw missing value is retained, alongside a separately flagged derived value of zero from total minus overnight arrivals. Eight other dates have one-arrival type-

sum discrepancies, which remain unadjusted. All nine flagged dates precede 2023 and do not affect the main regression sample.

We aggregate 1,461 daily records from 2019 and 2023-2025 into complete calendar quarters, then join the 16 expenditure observations one-to-one by year and quarter. Every quarter matches and contains all expected dates. The daily rows supply count denominators; the spending unit remains the city-quarter. Extraction distinguishes single-quarter, cumulative and annual columns, checks older 2019 layouts and changed 2025 column order, and records locations for 96 quarterly indicators.

Official spending per visit and values derived as published expenditure divided by arrivals are stored separately. Across 48 checks, the maximum absolute relative differences are 0.0211% overall, 0.0148% for overnight visitors and 0.0797% for same-day visitors. These small differences support matching units and periods, not an unbiased survey. Quarterly spending sums are MOP 1 million above the official annual value in 2019 and MOP 1 million below it in 2024. Both versions remain available.

Experiment 1 Arrival distributions and composition

For the fixed periods 2017-2019, 2020-2022 and 2023-2025, we calculate observed days, means, medians, IQR and P90, and compare the 2019 and 2025 empirical cumulative distribution functions (ECDFs). Means and quantiles distinguish total recovery from changes in typical and upper-tail days. A high-arrival threshold is a statistical definition, not an official capacity or congestion standard. Descriptive statistics and ECDFs follow Lecture 04 and Tutorial 07-01 (University of Macau, 2026c, 2026h).

Comparisons are stratified by month, weekday, and their combination. Monthly plots and the year-by-month heatmap use daily means; weekday plots use medians. All genuine high and low values remain. IQR describes spread and is not a deletion rule. These strata do not align lunar holidays, weather or events, remove all confounding, or justify default confidence bands that assume independent days.

The annual overnight share is annual overnight arrivals divided by annual total arrivals, not the unweighted mean of daily shares. Each source group is compared by counts, shares and absolute change from 2019 to 2025. Group changes must add to total change. Counts and percentage points help interpret opposing changes when the net change is small.

Totals, components, ratios and denominators are mathematically related, so overall and stratified results are reported together. We do not calculate correlation coefficients, test a Simpson reversal or identify behavioural mechanisms. The taught ideas of stratification and confounding guide the interpretation (University of Macau, 2026g).

Experiment 2 Calendar regression

The target is the raw daily total. Candidates are a training-period weekday-mean baseline, OLS with month and weekday, OLS adding a linear date trend, and Ridge with the same trend features. Month and weekday use one-hot encoding with a reference category dropped. Trend is days since 1 January 2023. There are no interactions, holiday, weather or lag features, or target transformations. Same-day source and visitor-type counts are excluded because they sum to the target. OLS and preprocessing follow Module 04 (University of Macau, 2026k).

Stage	Fixed design
Initial training	All 365 days in 2023
Validation	Each 2024 quarter is one validation window; training uses only earlier dates in 2023-2024

Stage	Fixed design
Ridge candidates	Alpha = 0.01, 0.1, 1, 10, 100, 1000; selected by mean fold MSE
Final fitting	Refit on all 731 days in 2023-2024 after selection
Final test	All 365 days in 2025; no changes to features or parameters based on test results

Encoding and scaling are fitted inside each training fold through a Pipeline. StandardScaler affects only the continuous trend; month and weekday dummies stay at 0/1. GridSearchCV uses explicit time indices. Alpha and the candidate model are chosen by the equally weighted mean of four 2024 fold MSEs; validation RMSE is the square root of that mean MSE. The expanding windows apply taught validation principles to ordered data; they are not a new specialist time-series model or a claimed copy of a course example. Ridge and model comparison follow Module 05 (University of Macau, 2026l, 2026m).

Primary evaluation uses MSE, RMSE and R^2; selection uses 2024 validation MSE only. MAE is an auxiliary description of mean absolute residuals, applying the taught ideas of means and residuals, rather than a separately claimed Module 05 topic. RMSE gives more weight to large errors. R^2 uses the observed test-period mean as an evaluation reference, can be negative, and is not prediction accuracy. Residuals equal actual minus estimated arrivals, so positive values mean underestimation. Errors are examined by month, weekday and observed arrivals above the 2023-2024 training P90. MSE and RMSE are also covered in Tutorial 10-02 (University of Macau, 2026i).

The 2025 data do not enter fitting, alpha search or selection scores, but they were viewed in the earlier descriptive analysis. This is therefore a historical holdout, not a fully blind test. Alpha search and candidate comparison share the 2024 folds; there is no nested validation. Publication latency is unknown. We do not establish real-time or future performance, test residual autocorrelation, or test coefficient significance.

Experiment 3 Linking arrivals and expenditure

The 16 quarters extend the spending question in Assignment 2. We compare arrivals, overnight shares, non-gaming expenditure and spending per visit by visitor type. Each recovery index uses the corresponding 2019 quarter as 100. Counts, money, per-visit amounts and shares retain clear units; raw values with incompatible units are not placed on the same axis.

Official annual spending per visit comes from annual publication tables, not an average of four quarterly values. Official and derived values remain separate. Visitor-type weights and within-type means help explain the overall mean. The weighted-mean relationship is approximate for rounded published values; we do not run a fixed-composition scenario or a factor-contribution decomposition.

All comparisons use nominal MOP, so changes may include prices and purchase mix. No tourism price-index deflation, Shapley decomposition or quarterly spending forecast is implemented. The 16 observations support description and consistency checks, not claims about real purchase volumes, causal contributions or future expenditure.

Links to course methods

Method	Course basis	Application
Wrangling, joins, groups and pivots	Assignment 01 Module 02; Tutorial 04-01, pp. 30-42	Dates, missing values, units, sums and one-to-one joins

Method	Course basis	Application
Descriptive statistics and empirical distributions	Lecture 04, pp. 19-56; Tutorial 07, pp. 7-18	Mean, median, IQR, P90 and ECDF
Proportions, weighted means and stratification	Lecture 04, pp. 20-26; Tutorial 05-02, pp. 3-10, 43-49	Type shares, within-group comparisons and confounding limits
Multiple OLS and Pipeline	Assignment 01 Module 04	Category encoding, trend scaling and calendar regression
Ridge and model evaluation	Modules 04/05 and Ridge page; Tutorial 10-02, pp. 7-9	Validation, MSE/RMSE/R^2 and residuals; auxiliary MAE
Matplotlib plots	Lecture 06, pp. 8-84, 91, 95; Lecture 05, pp. 37-40	Lines, distributions, heatmap, bars and error plots

Sampling distributions, confidence intervals, bootstrap and hypothesis tests are also taught. However, daily records are ordered and span different conditions, while expenditure is available only as published aggregates. We describe the finite historical periods directly rather than treating the records as independent random samples for ordinary t tests, random permutations or day-level bootstrap. Explaining this choice and its assumptions supports the course learning outcomes (University of Macau, 2026a, 2026f).

Project tasks and deliverables

The proposed workflow combines source verification, data preparation, distribution and composition analysis, quarterly reconciliation, and evaluation of four calendar models. The current implementation makes this scope feasible for six members within four weeks. We will verify and refine the existing evidence rather than treat completed experiments as future work. Any correction will be documented and propagated to the notebooks, result tables and report.

The existing experiment package contains two notebooks with 16 executed code cells, 24 CSV result tables, 13 figures, two Python analysis scripts, fixed inputs and source information. The planned final outputs are a verified experiment package, a concise report answering all four RQs, and presentation materials explaining the strongest findings and limitations. The notebooks explain each step, its course basis and its relation to the proposal. No dashboard is required for this scope.

Matplotlib guides the final line, distribution, bar, heatmap and error plots (University of Macau, 2026d, 2026e). We will retain limited model fit, high-arrival errors and differences across spending benchmarks as findings. Research depth comes from connecting compatible observation levels, testing arithmetic consistency, comparing conditional distributions and explaining changing model rankings without exceeding the taught methods.

Work step	Tools or guides	Expected output
1 Verify inputs and definitions	Pandas; DSEC definitions and source locations; Module 02	Checked daily and quarterly tables, data dictionary and anomaly log
2 Review arrival patterns and composition	Pandas groupby/pivot; Lecture 04; Tutorial 07; Matplotlib	RQ1-RQ2 tables, ECDFs, calendar comparisons, shares and source changes

Work step	Tools or guides	Expected output
3 Reproduce model comparison	scikit-learn Pipeline and GridSearchCV; Modules 04/05; Tutorial 10-02	RQ3 validation folds, alpha choice, all four test metrics and error diagnostics
4 Reconcile arrivals and spending	Pandas merge; DSEC quarterly and annual tables; weighted means	RQ4 recovery indices, visitor-type comparisons and rounding checks
5 Review interpretation and visuals	Course notes; notebook explanations; common chart checklist	Claims matched to evidence, readable figures and explicit limitations
6 Assemble and verify final delivery	Jupyter; requirements.txt; README; peer review checklist	Reproducible package, final report, presentation and submission copies

Success criteria

Questions and evidence. All four RQs must have a named result table or figure and a conclusion supported at the displayed precision. Every conclusion must state its period, observation unit and denominator; spending claims must state the nominal basis. SONG XUHUIYANG will check a claim-to-evidence list before release.

Data and traceability. There must be 3,287 unique daily dates, 17 source groups per date, and 16 complete matched quarters. Source-group counts must sum to daily totals. The one missing same-day value, eight one-arrival discrepancies and annual rounding differences must remain documented rather than silently removed. Official and derived spending per visit must remain separate.

Models and evaluation. Four chronological validation folds must use only earlier training dates. Preprocessing must be fitted on each fold's training dates only. Alpha and model selection must use 2024 validation MSE, with all four 2025 test results retained. Negative R^2 and high-arrival errors must be reported. A high R^2 is not a pass condition; an honest finding of limited performance is acceptable.

Charts and reproducibility. Both notebooks must run from a fresh kernel without errors using the documented environment and fixed data. The 24 CSV tables and 13 figures must regenerate, and reported numbers must agree with the tables at displayed precision. Every figure must have readable English labels, units, a clear comparison baseline and source information; comparable measures must use comparable scales. SONG will record the checks, and the team will close all material inconsistencies before release.

Team collaboration

The roles below follow Team 05's agreed work plan. HANG XIJI and GUO XIAOYU jointly hold the analyst and developer roles and are responsible for both notebooks. Responsibilities and review handoffs make each member accountable for the remaining work.

Member and role	Main responsibility	Handoff and peer check
HANG XIJI Coordinator; analyst/developer	Coordinate meetings, decisions and integration; jointly maintain both notebooks, data preparation and model evaluation with GUO.	Cross-review technical changes with GUO and hand both notebooks and RQ1-RQ4 evidence to SONG for quality checks.

Member and role	Main responsibility	Handoff and peer check
GUO XIAOYU Analyst/developer	Develop the research questions, data-source checks, tasks and timeline; jointly maintain both notebooks and interpretations with HANG.	Cross-review calculations and definitions with HANG; provide checked tables, figures and explanations to the reporters and SONG.
XU CHENYANG Scribe	Record minutes, decisions and changes; maintain file organization and format the proposal and report.	Reconcile document versions and references with HO; preserve the decision and change log.
HO IOK LAM Liaison; reporter	Check instructor requirements and relay feedback; lead the report narrative and presentation script.	Match claims to evidence with both analysts; obtain SONG's compliance review and coordinate final submission.
WU XIAQIN (KARENDA) Progress tracker; reporter	Maintain milestones, dependencies and risk log; integrate slides and organize rehearsal.	Check completion with HANG; review report-slide consistency with HO and track individual submissions.
SONG XUHUIYANG Quality reviewer	Independently check calculations, reproducibility, plots, citations and requirement compliance.	Record issues for the relevant owner, verify corrections and issue a final review checklist to all six members.

We will hold one online meeting each week and use the shared repository and task board between meetings. HANG will circulate the agenda; XU will record decisions, owners and due dates; WU will update progress and risks. Members will report blockers early and review one another's work constructively. Major scope and interpretation decisions will be made jointly. HO will send only team-approved questions or updates to the instructor and share the response with everyone. All six members will review the final report and presentation; each will complete any individually required submission.

Four week schedule

The schedule begins on the proposal submission date, 5 November 2026, and finishes on 2 December 2026: four consecutive seven-day blocks. The experiments and preliminary results already exist. The remaining work is verification, interpretation, report and presentation preparation, review and final delivery.

Period	Activities and responsible members	Milestone and acceptance evidence
Week 1 5-11 November	All members submit the proposal on 5 November. HANG and GUO freeze inputs and reproduce both notebooks; XU records versions; SONG checks data quality. HO confirms requirements; WU opens the progress log.	Verified data inventory, successful notebook runs and an agreed list of issues by 11 November.
Week 2 12-18 November	HANG and GUO cross-check daily and spending analyses. SONG checks all model scores, calendar comparisons, annual benchmarks and rounding differences. XU records corrections; HO and WU draft the evidence narrative.	All four RQs mapped to checked tables and figures; material numerical issues resolved by 18 November.

Period	Activities and responsible members	Milestone and acceptance evidence
Week 3 19-25 November	HO and WU integrate the report, slides and script. Analysts check interpretations; XU formats references and files; SONG reviews clarity and compliance. All six members rehearse and explain findings.	Complete report and presentation drafts, rehearsal notes and a short correction list by 25 November.
Week 4 26 November-2 December	Owners resolve review comments. SONG reruns the final checklist; XU assembles the release; HO and WU coordinate final reporting and submission; HANG confirms team agreement. Every member checks their required submission.	Final report, presentation, reproducible package and all required submissions completed by 2 December.

If a rerun or source check reveals a discrepancy, the responsible analyst will correct it and SONG will recheck the affected outputs. WU will record the impact on the schedule. We will preserve the four RQs and taught methods, prioritize correctness over optional presentation features, and finish within the four-week window.

Preliminary findings and feasibility

The following results come from the existing executed notebooks. They establish data availability, analytical feasibility and the value of the proposed questions. The scheduled team review will verify their reproduction and interpretation; these figures are not promised future outcomes.

RQ1 Total recovery and changing composition

Total arrivals increased by 663,179, or 1.68%, from 2019 to 2025. Overnight arrivals fell and same-day arrivals rose. The overnight share declined from 47.28% to 41.29%, a fall of 5.99 percentage points. This is a change in aggregate composition, not a longitudinal finding about individual visitors.

Measure	2019	2025
Total arrivals	39,406,181	40,069,360
Overnight arrivals	18,632,699	16,544,417
Same-day arrivals	20,773,482	23,524,943
Overnight share	47.28%	41.29%

Mainland China increased by 1,093,945 arrivals and its share rose from 70.86% to 72.42%. The Republic of Korea, Japan and Hong Kong declined by 195,388, 136,328 and 53,512 arrivals, respectively. The Mainland increase equals 164.95% of the overall net gain because other declines offset it. This is an arithmetic contribution to net change, not a visitor share or causal effect.

Excluding flagged type dates changes the 2019 overnight share by only -0.0114 percentage points and does not change 2025. This sensitivity comparison removes whole dates, not just their one-arrival discrepancies. The primary result retains all dates.

RQ2 Typical days and upper arrivals recover differently

Both years contain 365 days. In 2025, the mean was slightly higher but the median slightly lower. IQR and P90 were higher, while the maximum was lower. Total growth does not mean that all dates recovered together or that every peak increased.

Daily statistic	2019	2025
Mean	107,962.14	109,779.07
Median	104,695	103,252
IQR	27,163	34,032
P90	138,130.0	148,850.2
Maximum	226,326	221,857

Changes also differ by calendar stratum. August daily means rose from 116,874.71 to 136,097.87, while February means fell from 126,632.18 to 112,399.43. Saturday medians rose from 129,615 to 138,702; Tuesday medians fell from 93,965 to 92,336.5. Moving holidays and other conditions still differ within these strata, so the comparisons remain descriptive.

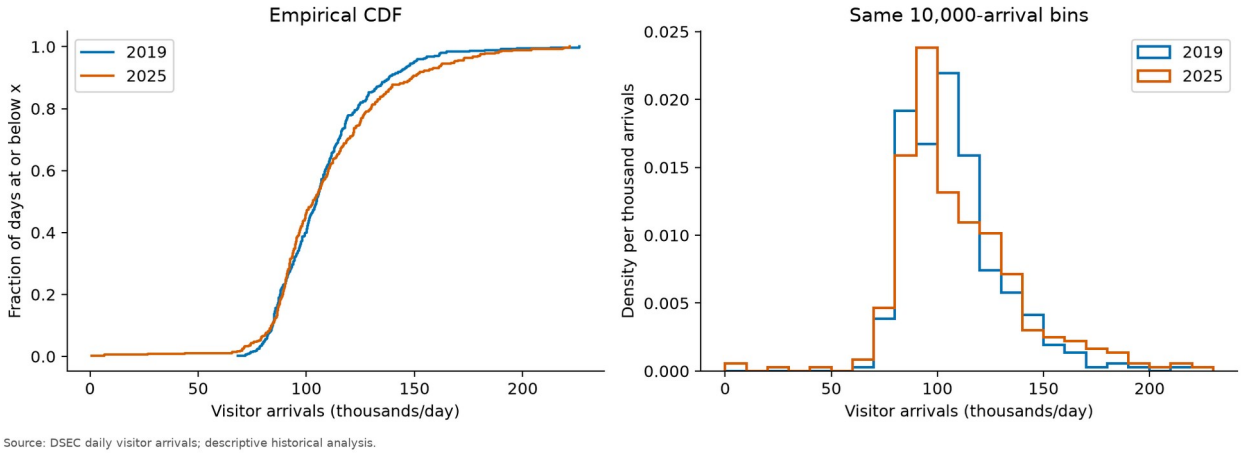


Figure 1. Daily arrival distributions. The horizontal axes show thousands of arrivals per day. The left panel shows the proportion of dates at or below each value; the right uses density with common bin widths. Source: DSEC; generated by analysis.ipynb.

RQ3 Model performance depends on the evaluation period

The four 2024 validation folds select Ridge with alpha = 100. The table retains all candidates. Validation RMSE is the square root of mean fold MSE. RMSE and MAE are in arrivals per day; R^2 is unitless.

Model	Validation RMSE	Test RMSE	Test MAE	Test R^2
Weekday mean	26,133.08	33,832.00	25,585.71	-0.3491
OLS calendar	30,876.38	33,218.97	25,113.43	-0.3007
OLS calendar + trend	73,753.37	23,916.19	16,728.15	0.3258
Ridge calendar + trend	24,820.82	25,818.77	18,767.15	0.2143

Ridge performs best in validation, but OLS with a trend has lower 2025 test error. Ridge therefore does not dominate every comparison. The high mean validation error for OLS with a trend is driven mainly by its first fold. Rankings change with the training history and evaluation period. We retain the validation-selected Ridge and report the test comparison without using it to retune the model.

Ridge test R^2 is 0.2143: its squared error is about 21.43% lower than the reference based on the observed test mean. This is not prediction accuracy. Calendar variables leave much daily variation unexplained, so the results do not support precise peak alerts.

Actual 2025 arrivals	Days	RMSE	Mean residual
Above training P90	103	36,510.33	+27,053.09
At or below training P90	262	20,115.40	-14,061.04

The training P90 is 121,707 arrivals. Dates above it are underestimated by about 27,053 arrivals on average; other dates are overestimated by about 14,061. Opposing errors partly cancel in the overall mean. The groups use actual 2025 arrivals and are retrospective diagnostics, not a prospective classifier or an official capacity rule.

RQ4 Recovery depends on the measure and benchmark

With the corresponding 2019 quarter set to 100, 2025 first-half arrivals remain below baseline while nominal expenditure and official spending per visit are above it. All three exceed baseline in the second half, by different amounts. An index of 115.9 means 15.9% above baseline, not 115.9% growth.

2025 quarter	Arrivals index	Nominal spending index	Official per-visit index
Q1	95.2	115.9	121.7
Q2	94.3	116.1	123.2
Q3	105.4	134.1	127.3
Q4	113.0	134.7	119.2

Official annual non-gaming expenditure in 2025 is MOP 80,122 million, and official spending per visit is MOP 2,000. Annual comparisons below use official annual values. Calculations retain stored precision, while displayed figures are rounded.

Measure	2025 vs 2019	2025 vs 2024
Arrivals	+1.68%	+14.72%
Nominal non-gaming expenditure	+25.04%	+6.32%
Official nominal spending per visit	+23.00%	-7.28%

Above the pre-pandemic benchmark and below the previous year can both describe spending per visit. The table uses official annual expenditure. Using sums of quarterly expenditure instead gives 2025 growth of 25.0382% against 2019 and 6.3232% against 2024, close to but distinct from 25.0402% and 6.3218% using official annual totals. Both definitions and the rounding differences are retained.

Overnight spending per visit exceeds same-day spending in all 16 quarters. In 2025 Q3, overnight visitors account for 41.68% of arrivals but 82.22% of expenditure; official spending per visit is MOP 3,846 versus MOP 595. The annual overnight arrival share falls from 45.93% in 2024 to 41.29% in 2025. Annual overnight spending per visit falls from MOP 3,884 to 3,827, while same-day spending rises from MOP 691 to 715. An overall decline does not require both type means to fall: weights and within-type means matter. We do not estimate their individual percentage contributions.

Limitations

Observation units and dependence. Daily coverage is complete for the selected historical period, but adjacent dates may be related and conditions differ across periods. Source records share dates and summation constraints; city-quarter expenditure is not respondent microdata. Record counts do not replace an independent sample size or justify ordinary iid significance calculations.

Survey and price basis. Expenditure is a quota-survey estimate; arrivals are administrative counts joined only by quarter. Without respondent records and survey variance information, we cannot analyse personal spending distributions or construct respondent confidence intervals. Nominal amounts are not deflated and do not measure changes in real purchasing power, tourism GDP, profit or welfare.

Definitions and quality. Missing type values, one-arrival discrepancies and million-MOP rounding differences remain visible. The wording of overnight definitions differs across historical publications, which should be disclosed without inferring a statistical break from wording alone. Arithmetic agreement cannot establish that official data have no measurement error.

Selection and extrapolation. The descriptive analysis viewed 2025, and alpha tuning and candidate comparisons share the 2024 folds. The design is neither fully blind nor nested. Models contain only additive month, weekday and linear trend terms; they omit holidays, weather, events, policy and transport supply. Residual autocorrelation has not been formally tested. Error peaks cannot automatically be attributed to particular events.

Interpretation and use. Source contributions are arithmetic changes; type means and stratified comparisons are associations. There is no random assignment, causal identification, fixed-composition counterfactual or Shapley decomposition. Publication latency is unknown, and historical 2025 performance does not establish future or real-time reliability.

Files and reproducibility

analysis.ipynb covers daily distributions, annual and stratified tables, source changes, validation, test metrics and residuals. spending_analysis.ipynb covers quarterly recovery indices, visitor-type shares, official-versus-derived spending per visit and annual comparisons against both benchmarks.

The data folder contains fixed inputs and measurement notes; results contains 24 machine-readable CSV tables; figures contains 13 plots. The two Python scripts and requirements.txt support reruns, and protocol.txt records the design. Official source URLs remain in the data and documentation. The final team review will use README.txt to reproduce the package and reconcile each reported claim with its output. Any change to inputs, code or definitions will require the corresponding explanations and report figures to be rechecked.

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